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Keith A Knurr[1,2], Daniel G Cobian[1,2], Stephanie A Kliethermes[1,2], Mikel R Joachim[1,2], and Bryan C Heiderscheit[1,2,3]

[1]Department of Orthopedics & Rehabilitation, University of Wisconsin-Madison, Madison, WI;
[2]Badger Athletic Performance Program, University of Wisconsin-Madison, Madison, WI;
[3]Department of Biomedical Engineering, University of Wisconsin-Madison, Madison, WI

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Keith A Knurr[1,2], Daniel G Cobian[1,2], Stephanie A Kliethermes[1,2], Mikel R Joachim[1,2], and Bryan C Heiderscheit[1,2,3]

[1]Department of Orthopedics & Rehabilitation, University of Wisconsin-Madison, Madison, WI;
[2]Badger Athletic Performance Program, University of Wisconsin-Madison, Madison, WI;
[3]Department of Biomedical Engineering, University of Wisconsin-Madison, Madison, WI

Address for correspondence: Keith A. Knurr, PhD, DPT, SCS, Department of Orthopedics and Rehabilitation, University of Wisconsin School of Medicine and Public Health, 1685 Highland Avenue, Madison, WI 53705-2281; E-mail: knurr@ortho.wisc.edu; Office: (608) 265-8122

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ABSTRACT

Purpose: Following anterior cruciate ligament reconstruction (ACLR), individuals present with significant and persistent deficits in surgical knee biomechanics during running. Little is known regarding the role of the hip and ankle in compensating for these knee-specific deficits. The purpose of this study was to characterize how limb (surgical and non-surgical), time post-operatively (3.5-7 months [EARLY], 8-13 months [LATE]), and running speed influence lower extremity joint-specific and total work during running post-ACLR. **Methods:** Fifty-six Division I collegiate athletes post-ACLR (EARLY: n=41, LATE: n=42, Both: n=27) completed running analyses at 2.68, 2.95, 3.35, 3.80 and 4.47 m/s or up to their maximum comfortable speed. Linear mixed effects models assessed the influence of limb, speed, time post-ACLR, and their interactions on hip, knee, ankle, and total negative and positive work. **Results:** Surgical limb hip positive work was greater (0.044 J/kg [0.015, 0.074], p<0.001) than the non-surgical limb EARLY. Surgical limb ankle negative (-0.054 J/kg [-0.093, -0.015], p=0.003) and positive work (-0.115 J/kg [-0.168, -0.063], p<0.001) were less than the non-surgical limb EARLY. Surgical limb knee negative (EARLY: -0.339 J/kg [-0.382, -0.296], p<0.001; LATE: -0.222 J/kg [-0.265, -0.180], p<0.001) and positive work (EARLY: -0.214 J/kg [-0.246, -0.182], p<0.001; LATE: -0.142 J/kg [-0.174, -0.110], p<0.001) were less than the non-surgical limb. Total negative and positive work followed the same pattern as the knee. Conclusions: Athletes post-ACLR exhibited greater hip positive work and lesser ankle positive and negative work during running of the surgical limb EARLY, that resolved LATE. Both knee and total negative and positive work of the surgical limb were substantially less than the non-surgical limb, which improved from EARLY to LATE, but between-limb asymmetries remained.

Key Words: ACL, BIOMECHANICS, CONTRIBUTION, JOINT, KNEE

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INTRODUCTION

Following anterior cruciate ligament reconstruction (ACLR), athletes often exhibit alterations in knee joint biomechanics during running.(1) These alterations commonly include reduced surgical knee peak flexion angles, flexion excursions, and peak extensor moments compared to the non-surgical limb,(1–3) healthy controls,(1,4–6) and their own preinjury biomechanics.(7) Altered movement biomechanics following ACLR persist long-term(8) and are thought to contribute to reduced athletic performance,(9,10) increased susceptibility to re-injury,(11,12) and development of post-traumatic osteoarthritis (PTOA).(13–15) As running is an integral component of most sports and a key milestone in the rehabilitation process(16), it is an important task to monitor post-operatively. While considerable attention has been dedicated to knee-specific biomechanics, understanding of the broader changes across lower extremity joints during running is limited. Previous investigations suggest that athletes following ACLR may demonstrate compensatory strategies at the ankle,(2,7) hip,(2,8) and/or the trunk.(17,18) It is unknown if these inter-joint compensatory strategies follow a similar recovery pattern as the knee-specific biomechanics.

Assessing hip, knee, and ankle joint work allows for an intuitive interpretation of inter-joint compensation during running, which can be separated into positive and negative work. In general, during running, joint negative work occurs when muscles contract eccentrically to absorb energy during the loading response phase, while positive work occurs when muscles contract concentrically to generate force during the propulsive phase.(19) Knee negative work is an important biomechanical variable to assess in athletes following ACLR, as it largely reflects the eccentric contributions of the quadriceps during the loading response of running.(20,21) Compared to other commonly reported knee kinetics, such as peak knee extensor moment, knee negative

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work has shown more pronounced surgical-limb deficits compared to preinjury(7) and the non-surgical limb.(22) Our prior exploratory work suggests that the reduction in knee negative work of the surgical limb contributes to an overall reduction in total negative work, with potential minor changes at the ankle.(23)

When assessing running biomechanics post-ACLR it is important to consider both the time post-operatively as well as the running speed.(23) Early post-operatively, athletes may not be able to generate the knee joint torque necessary for running symmetrically due to profound quadriceps performance impairments.(24) As such, individuals may use compensatory strategies, such as redistributing work to the ankle or hip to successfully complete the task.(25) With an increase in running speed, knee biomechanics of both the surgical and non-surgical limb increase similarly in athletes post-ACLR,(23) but it is unknown if hip and ankle joint biomechanics follow a similar pattern, particularly at different time points post-operatively. For example, athletes may disproportionately increase the work done at the ankle and/or the hip of the surgical limb relative to the knee in response to an increase in running speed earlier in the rehabilitation process when knee impairments are most profound, while later in the rehab process the same magnitude of compensation may not be required. Better understanding of the influence of speed and time post-ACLR on joint-specific and total work performed may provide new insight into strategies for mitigating aberrant running biomechanics post-ACLR.

Despite the prominence of knee deficits, there is a knowledge gap regarding the potential compensatory role the hip and ankle may have during clinically relevant stages of recovery post-ACLR. Therefore, the purpose of this study was to characterize how limb (surgical and non-surgical), time post-operatively (3.5-7 months [EARLY], 8-13 months [LATE]), and running speed influence lower extremity joint-specific and total work during running in NCAA Division I

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collegiate athletes post-ACLR. With respect to time, we hypothesized that EARLY post-ACLR, surgical limb hip and ankle positive and negative work will be increased relative to the non-surgical limb as a compensatory strategy for significantly reduced knee negative and positive work of the surgical limb. At LATE post-ACLR, we hypothesized that a similar compensatory strategy of increased hip and ankle negative and positive work would be observed, albeit to a lesser extent than EARLY post-ACLR. We also hypothesized that total negative and positive work would be reduced on the surgical limb compared to the non-surgical limb at both time points, but would improve from EARLY to LATE. With respect to running speed, we hypothesized that hip, knee, ankle, and total negative and positive work would increase similarly between limbs with an increase in speed as observed in healthy individuals.(26,27)

METHODS

Participants and Study Design

This project expands on a previous prospective study that exclusively investigated the effects of running speed and time post-ACLR on knee specific biomechanics during running in Division I collegiate athletes at the University of Wisconsin-Madison.(23) The university's Health Sciences Institutional Review Board approved this records review. Given that these data came from a sample of convenience, no power analysis was conducted. To be included in the dataset, athletes had to meet the following criteria: 1) had a primary ACLR, 2) had no lower extremity surgery since the ACLR or 12 months prior to the ACLR, 3) completed a running analysis collection between 3.5-7 months and/or 8-13 months post-ACLR (collections between 7.0-8.0 months post-ACLR were excluded to create a temporal separation), and 4) had running data for at least 2 of the following speeds: 2.68m/s, 2.95m/s, 3.35m/s, 3.80m/s, and 4.47m/s.

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Running Analysis

A detailed description of running data acquisition and analysis procedures have been previously reported.(23) In brief, marker-based, three-dimensional motion analysis using an 8-camera system (Motion Analysis Corporation, Santa Rosa, CA, USA) was performed while athletes post-ACLR ran on an instrumented treadmill (Bertec Corporation, Columbus, OH, USA). First, 52 reflective markers were placed on the athlete and a static standing posture and hip circumduction tasks were recorded to identify joint centers and for model scaling. Following a walking acclimation period (at least 2 minutes), speed was incrementally increased from 2.68 m/s to 4.47 m/s (10 min/mile to 6 min/mile, at 1 min/mile increments) or until the athlete reached their self-reported maximum comfortable speed. To ensure all strides were indeed running strides, a flight phase was confirmed between each foot contact based on ground reaction force (GRF) time-series. Kinematics (200 Hz) and GRF (2000 Hz) were synchronously recorded for 15 seconds at each speed following at least 30 seconds of acclimatization.

To model the body, a 14-segment, 31 degrees-of-freedom articular linkage, scaled to the athlete's height, mass, and segment lengths was used.(28) Kinematic and GRF data were low-pass filtered using a bi-directional, 4th-order Butterworth filter with a cutoff frequency of 12 Hz, which were subsequently used for segment-by-segment inverse dynamics analysis to calculate joint moments. Hip, knee, and ankle joint powers were calculated as the product of joint moment and angular velocity, and mechanical work was computed by integrating the positive and negative portions of the power curves. Separately, raw GRF data were filtered at 50 Hz and used only for stance phase identification using a 50 N threshold for vertical GRF.(29) Variables of interest included: hip, knee, ankle, and total negative work (area under the negative power curve during the loading response phase of the stance phase) and positive work (area under the positive power

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curve during the propulsion phase of the stance phase). Joint specific negative and positive work regions of interest are illustrated in Figure 1. Total negative work was computed as the sum of the hip, knee, and ankle work. All signal and data processing were conducted using MATLAB Release 2022b (The MathWorks Inc, Natick, MA, USA).

Statistical Methods

Means and standard deviations or frequencies and percentages were used to describe the athlete sample. Multivariable linear mixed effects models assessed the influence of speed, limb, and time post-ACLR on work performed during running. All 2-way interactions were assessed and if non-significant were removed from the model. For the independent variables, speed (5 levels), limb (2 levels), and time (2 levels) were treated as categorical variables. Athlete and limb were assigned as random effects with limb nested within athlete. Missing data was assumed to be missing at random. A sensitivity analysis has previously shown that this is a valid assumption given minimal difference between the full dataset and a reduced dataset of only athletes that achieved greater than 3.80 m/s.(23) For all pairwise comparisons, Tukey-adjusted p-values were reported. Least square mean differences, associated 95% confidence intervals [lower bound, upper bound], and Cohen's D effect sizes (ES) are reported. Statistical analyses were performed using R (version 4.4.0; R Core Team (2024)), and significance was assessed at $\alpha = 0.05$.

RESULTS

Fifty-six Division I collegiate athletes were included in this study (41 athletes EARLY; 42 athletes LATE; Table 1). Twenty-seven athletes had running analyses at both time points. A summary of all final model estimates, interaction and main effects, are provided in Table 2. All

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least-square means and 95% confidence intervals for hip, knee, ankle, and total negative and positive work are reported across speeds and time points for both the surgical and non-surgical limb in Supplemental Tables 1 and 2 (Supplemental Digital Content 1) and visualized in Figure 2. Figure 3 (negative work) and Figure 4 (positive work) highlight the between-limb differences in hip, knee, ankle, and total work as they related to overall work done by the lower extremities.

Hip Work

A significant interaction between limb and time was observed for hip negative ($p = 0.01$) and positive work ($p = 0.046$). Hip negative work of the surgical limb increased from EARLY to LATE (0.014 J/kg [0.005, 0.023], $p<0.001$, ES = 0.16), when accounting for running speed. At LATE, no between-limb difference in hip negative work was identified ($p =0.99$). For hip positive work, the surgical limb was greater than the non-surgical limb EARLY (0.044 J/kg [0.015, 0.074], $p<0.001$, ES = 0.42), but not LATE (0.022 J/kg [-0.007, 0.051], $p = 0.19$, ES = 0.22). The limb by speed interaction was not significant for hip positive work ($p = 0.61$) and negative work ($p = 0.76$), but there was a main effect for speed for hip positive work ($p<0.001$) and negative work ($p<0.001$), indicating both limbs increased similarly with an increase in running speed, when accounting for time post-operatively.

Knee Work

A significant interaction between limb and time was observed for knee negative ($p<0.001$) and positive work ($p<0.001$). Knee negative work of the surgical limb increased EARLY to LATE (0.097 J/kg [0.079, 0.114], $p<0.001$, ES = 0.59), while the non-surgical limb decreased (-0.020 J/kg [-0.037, -0.002], $p = 0.02$, ES = 0.12), when accounting for running speed. A similar pattern

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was observed for knee positive work in that the surgical limb increased from EARLY to LATE (0.059 J/kg [0.044, 0.073], p<0.001, ES = 0.43), but the non-surgical limb did not significantly change (-0.013 J/kg [-0.027, 0.001], p = 0.09, ES = 0.10), when accounting for running speed. Knee negative work of the surgical limb was less than the non-surgical limb both EARLY (0.339 J/kg [0.296, 0.382], p<0.001, ES = 2.54) and LATE (0.222 J/kg [0.180, 0.265], p<0.001, ES = 1.71), when accounting for running speed. Knee positive work of the surgical limb was less than the non-surgical limb EARLY (0.214 J/kg [0.182, 0.246], p<0.001, ES = 2.10) and LATE (0.142 J/kg [0.110, 0.174], p<0.001, ES = 1.43), when accounting for running speed. The limb by speed interaction was not significant for knee positive work (p = 0.31) and negative work (p = 0.38), but there was a main effect for speed for knee positive work (p<0.001) and negative work (p<0.001), indicating both limbs increased similarly with an increase in running speed, when accounting for time post-operatively.

Ankle Work

A significant interaction between limb and time was observed for ankle negative (p = 0.04) and positive work (p<0.001). Ankle negative work of the surgical limb increased EARLY to LATE (0.039 J/kg [0.018, 0.060], p<0.001, ES = 0.19), when accounting for running speed. Ankle positive work done by the surgical limb increased from EARLY to LATE (0.096 J/kg [0.068, 0.123], p<0.001, ES = 0.37), as well as the non-surgical limb (0.029 J/kg [0.002, 0.056], p = 0.03, ES = 0.11), when accounting for running speed. Ankle negative work of the surgical limb was less than the non-surgical limb EARLY (0.054 J/kg [0.015, 0.093], p = 0.003, ES = 0.42), but not LATE 0.030 J/kg [-0.009, 0.068], p = 0.18, ES = 0.24), when accounting for running speed. Ankle positive work of the surgical limb was less than the non-surgical limb EARLY (0.115 J/kg [0.063,

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0.168], p<0.001, ES = 0.67), but not LATE (0.049 J/kg [-0.003, 0.101], p=0.07, ES = 0.29), when accounting for running speed. The limb by speed interaction was not significant for ankle positive work (p = 0.67) and negative work (p = 0.87), but there was a main effect for speed for ankle positive work (p<0.001) and negative work (p<0.001), indicating both limbs increased similarly with an increase in running speed, when accounting for time post-operatively.

Total Work

A significant interaction between limb and time was observed for total negative (p<0.001) and positive work (p<0.001). Total negative work done by the surgical limb increased EARLY to LATE (0.151 J/kg [0.121, 0.209], p<0.001, ES = 0.53), when accounting for running speed. Total positive work done by the surgical limb increased from EARLY to LATE (0.145 J/kg [0.103, 0.186], p<0.001, ES = 0.37), when accounting for running speed. Total negative work of the surgical limb was less than the non-surgical limb EARLY (0.424 J/kg [0.359, 0.488], p<0.001, ES = 2.06) and LATE (0.272 J/kg [0.209, 0.336], p<0.001, ES = 1.32), when accounting for running speed. Total positive work of the surgical limb was less than the non-surgical limb EARLY (0.284 J/kg [0.206, 0.363], p<0.001, ES = 1.10) and LATE (0.169 J/kg [0.091, 0.246], p<0.001, ES = 0.68), when accounting for running speed. The limb by speed interaction was not significant for total positive work (p = 0.38) and negative work (p = 0.96), but there was a main effect for speed for total positive work (p<0.001) and negative work (p<0.001), indicating both limbs increased similarly with an increase in running speed, when accounting for time post-operatively.

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DISCUSSION

This study sought to describe how surgical and non-surgical limb hip, knee, ankle, and total negative and positive work change with running speed and time post-ACLR in NCAA Division I collegiate athletes. Contrary to our initial hypothesis, surgical limb ankle negative and positive work were less than the non-surgical limb at the EARLY timepoint. Only hip positive work of the surgical limb was greater compared to the non-surgical limb EARLY but not LATE post-ACLR. As expected, the work done by the surgical knee joint was significantly less than the non-surgical limb, and although this improved from EARLY to LATE, substantial deficits remained at LATE. As we hypothesized, total negative and positive work of the surgical limb were significantly less than the non-surgical limb. Surgical limb total work increased from EARLY to LATE, but substantial deficits remained at LATE. In our previous study we observed that knee negative work of both limbs increased similarly with an increase in running speed.(23) Consistent with this prior work, in the current study we observed a similar increase in all joint-specific and total work metrics of both limbs in response to an increase in running speed.

Increasing running speed does not appear to increase the between-limb asymmetries and may be a viable option to increase the amount of negative and positive joint and total work performed. Our findings suggest that increasing running speeds post-ACLR may promote increased surgical limb work, but this decision should be made with careful consideration of patient-specific factors to avoid increasing risk of re-injury before graft maturation is achieved. In our clinical opinion, faster, steady-state running should be introduced in short intervals with a gradual increase in speed over time, while closely monitoring the athlete's response to the increase in running speed. It is important to highlight that this study investigated steady-state running at speeds up to 4.47 m/s

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(6:00 min/mile); therefore, it is unknown if these clinical implications carry forward to faster running speeds or rapid acceleration and deceleration tasks.

Compensatory movement strategies are a concern for athletes following ACLR, given the prevalence of PTOA.(30) Altered movement mechanics, characterized by reduced knee extensor demand and a less dynamic vertical ground reaction force profile, have been associated with poorer cartilage composition measures as early as 6 and 12 months post-ACLR.(15,31,32) Moreover, lower patellofemoral joint contact pressures during walking at 6 month post-ACLR were associated with worse cartilage health at 3-years.(33) Patients with radiographic PTOA at 5 years post-ACLR walked with lower knee adduction moments and medial compartment joint contact forces compared to those without radiographic PTOA.(34) Taken together, these studies and others(13,14,35), highlight that chronic underloading of the surgical limb is associated with poorer cartilage health in individuals post-ACLR, and suggest that restoring movement symmetry by increasing surgical limb loading has the potential to mitigate the onset and progression of knee PTOA. While prior research related to PTOA and movement mechanics has primarily focused on walking and jumping tasks, consistent surgical limb underloading (reduced extensor moments and knee flexion excursion) has been observed across tasks(36–39), including running.(1) These findings suggest that this knee underloading may contribute to the mechanical environment that promotes PTOA development.

We observed a 67% deficit in knee negative work EARLY and a 45% deficit LATE in the surgical limb compared to the non-surgical limb, which is similar to previous studies on running in athletes post-ACLR.[7,20] It is apparent that the negative work done by the surgical knee is substantially reduced and well outside of the normal asymmetries observed in healthy collegiate athletes (15.8%; 95% CI: [12.9, 18.7]).(40) It is important to highlight the relationship of

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quadriceps strength and rate of torque development (RTD) with movement mechanics across a variety of tasks.(6,41–45) Deficits in quadriceps strength and RTD are nearly ubiquitous following ACLR and are a key focus of rehabilitation.(46) However, despite athletes having near full recovery of quadriceps strength and rate of torque development, running asymmetries persist.(45) Positive work during running has been investigated less often than negative work in the context of ACLR, as the work done by the knee is substantially greater during the loading response phase(47) and reflects the eccentric contributions of the quadriceps.(20,21) However, this study demonstrates that athletes also have significant and persistent deficits in their production of positive work about the knee during the latter half of stance. Large deficits in knee kinematics and kinetics during the loading response phase suggests it will be challenging to generate large knee angular velocities and moments during the second half of stance, which are primary components of joint work.(23,48)

The only apparent compensation that may help to offset the substantial reduction in knee work performed was an increase in hip positive work that was observed EARLY post-ACLR. Similarly, previous research has demonstrated an increase in surgical limb hip extensor moments in comparison to the non-surgical limb and healthy controls.(2,8) This compensatory strategy is likely in response to the reduced knee negative work done during loading response, ultimately necessitating that the hip extensors perform more work to control the center of mass vertical displacement.(20) It is important to highlight that the increase in surgical limb hip positive work post-ACLR, particularly early in the rehabilitation process, is relatively minor (13.7%), making it an unlikely target for clinical interventions; however, it is plausible that resolving knee negative work deficits may also result in improved hip positive work symmetry.

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Following the pattern at the knee, we also observed reduced surgical limb ankle negative and positive work EARLY. However, no between-limb differences were detected for either positive or negative ankle work LATE. Again, this compensation may simply be a result of the large reduction in flexion angles and joint torques observed at the knee that drives similar impairments at the ankle. During squatting, within-limb, inter-joint compensations have been observed where hip extensor kinetics are increased to partially counter the deficits at the knee.(49) Interestingly, this strategy corresponded with no significant between-limb differences in peak GRF at 5 months post-ACLR. However, running is a different movement pattern and does not seem to present with a significant shift in inter-joint work, except for a relatively small increase in hip positive work EARLY (13.7%). Rather, a more global compensation occurs, which is reflected in the reduction of total negative and positive work done by the surgical limb. This is largely driven by the reduction in negative and positive work at the knee, but reduced work at the ankle was also observed EARLY. At LATE, no between-limb differences in positive or negative work were observed at the hip or ankle, but large asymmetries were still present at the knee resulting in reduced total work. Although not directly reported in this study, reduced center of mass vertical displacement during the stance phase of the surgical limb likely contributes to this persistent reduction in total joint work.(50) Taken together, this study supports that interventions aiming to restore normal running mechanics post-ACLR should attempt to increase surgical knee joint work and/or total surgical limb work.

Restoring movement symmetry during the rehabilitation process post-ACLR may lead to a reduction in risk and progression of PTOA,(13) but simply re-introducing running in a graded fashion based on symptomatology does not lead to improved symmetry during running.(51) Unfortunately, gait retraining strategies, particularly for running post-ACLR, are limited. Utilizing

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real-time biofeedback targeting knee kinetics(52) and ground reaction forces(53) may be viable options for modifying gait mechanics in individuals post-ACLR; however, these approaches require three-dimensional motion capture systems and a force instrumented treadmill or force plates, which limits their clinical feasibility. Decreasing running step rate to increase the demand on the knee extensors through cadence manipulation(54) may be a low tech, clinically feasible strategy for modifying gait in athletes post-ACLR, but further research is needed in a population of individuals post-ACLR and to explore the impacts on cartilage health or pain.

It is important to note that there are limitations to this current investigation. This study is a secondary analysis of a prior study and the data came from a database of routinely collected clinical and performance assessments of collegiate athletes. Consequently, not every athlete underwent testing at both specified times or ran at every designated speed. To address this, missing data was assumed to be missing at random and linear mixed effects models were used. To prioritize athlete safety and facilitate an incremental warm-up for higher speeds, running speed was not randomized. Rehabilitation was not controlled between athletes; however, athletes did complete rehabilitation in the same sports medicine facility, following similar post-operative rehabilitation guidelines. Findings from this study may have limited generalizability to athletes of different ages, graft types, and competition levels.

In conclusion, only hip positive work of the surgical limb during running increased compared to the non-surgical limb, which was present between 3.5-7 months post-ACLR in Division I collegiate athletes. Ankle negative and positive work were reduced on the surgical limb compared to the non-surgical limb between 3.5-7 months post-ACLR. Surgical limb knee joint and total negative and positive work during running were significantly reduced compared to the non-surgical limb, and although improvements were observed over time, substantial deficits remained

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8-13 months post-ACLR. All joint and total work of both limbs increased at a similar rate with an increase in running speed.

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FIGURE LEGENDS

Figure 1. Representative sample of hip, knee, and ankle joint negative (red) and positive work (blue) regions of interest during the stance phase of running. The dashed vertical line at 50% of the stance phase represents midstance as defined by the point where the anterior-posterior ground reaction force transitions from braking force to propulsive force. IC = initial contact; TO = toe off.

Figure 2. Least square means and 95% confidence intervals for negative work (A = hip, C = knee, E = ankle, G = total) and positive work (B = hip, D = knee, F = ankle, H = total) across running speeds for both limbs and time points. *, significant within-limb difference from 2.68 m/s (Tukey-adjusted p-values < 0.05) when accounting for time; ŧ, significant within-limb change over time (Tukey-adjusted p-values < 0.05) when accounting for speed; #, significant between-limb difference when accounting for speed (p-value < 0.05). EARLY, 3.5-7 months post-ACLR; LATE, 8-13 months post-ACLR.

Figure 3. Hip, knee, ankle, and total negative work as they related to overall work done by the lower extremities with respect to time post-ACLR and running speed. The red border signifies the surgical limb joint work as a proportion of total work done by both limbs. The center circle scales with the cumulative total work of both limbs. EARLY, 3.5-7 months post-ACLR; LATE, 8-13 months post-ACLR.

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Figure 4. Hip, knee, ankle, and total positive work as they related to overall work done by the lower extremities with respect to time post-ACLR and running speed. The red border signifies the surgical limb joint work as a proportion of total work done by both limbs. The center circle scales with the cumulative total work of both limbs. EARLY, 3.5-7 months post-ACLR; LATE, 8-13 months post-ACLR.

SUPPLEMENTAL DIGITAL CONTENT

SDC 1: Supplemental Digital Content 1.docx

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Figure 1

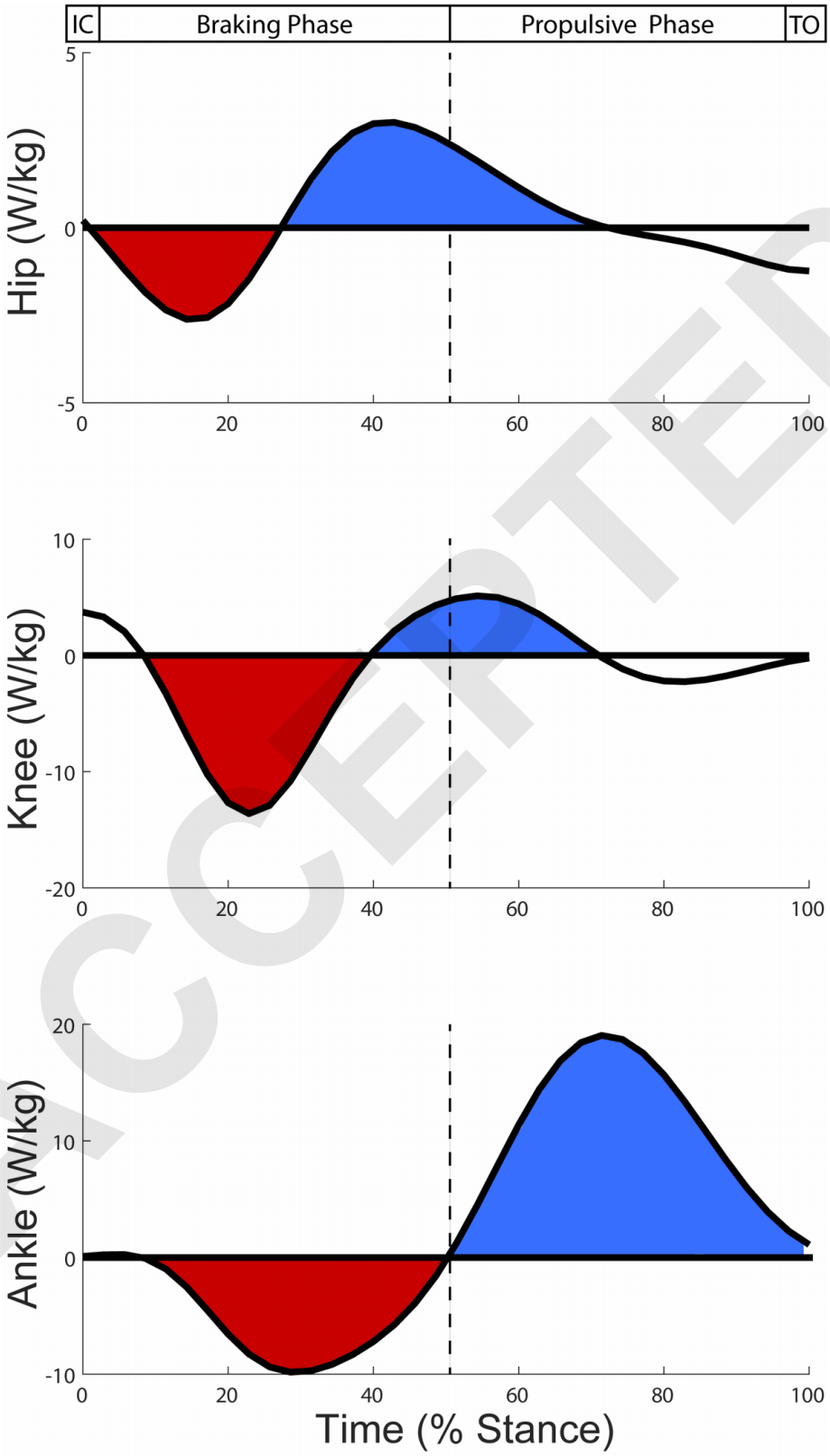

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Figure 2

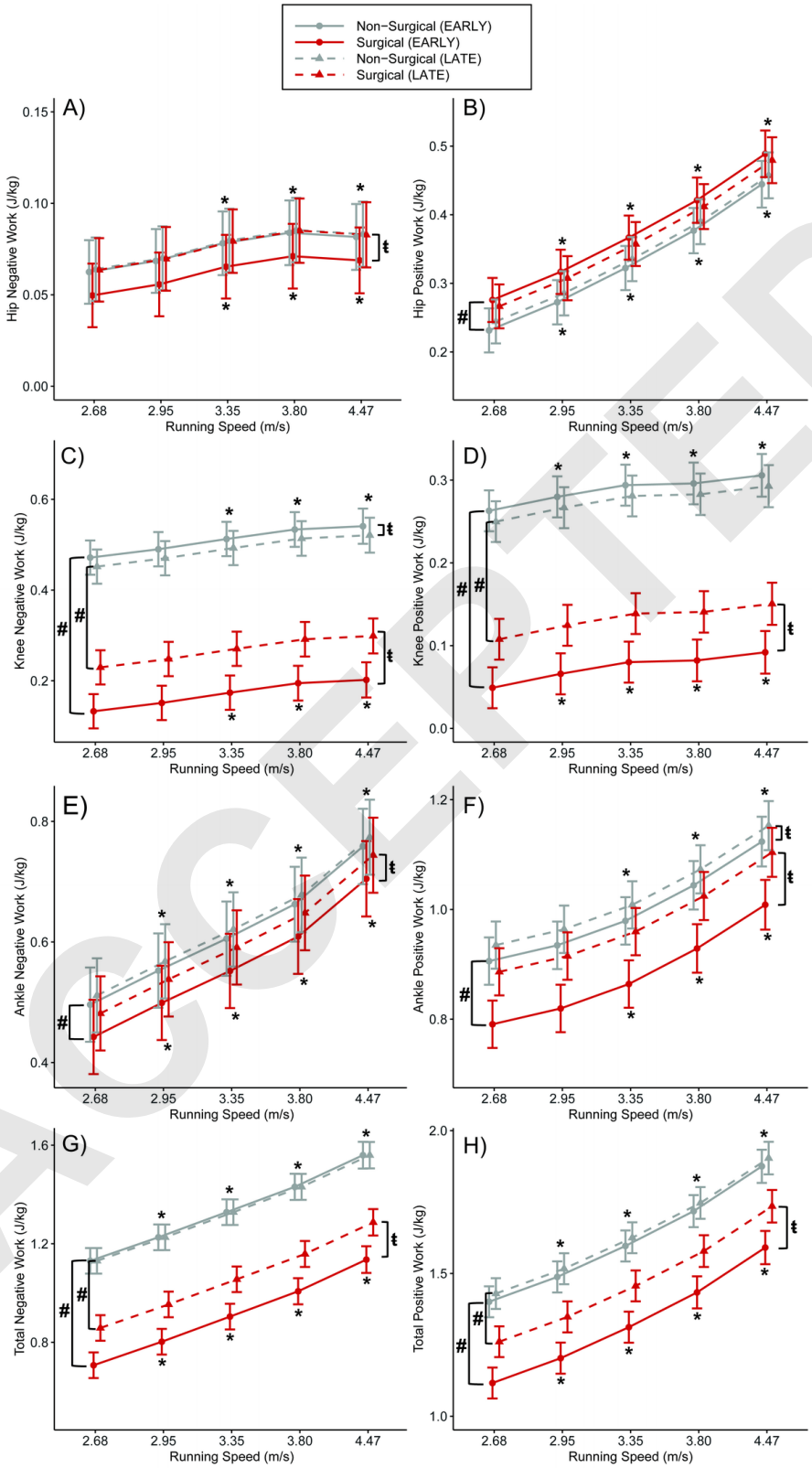

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Figure 3

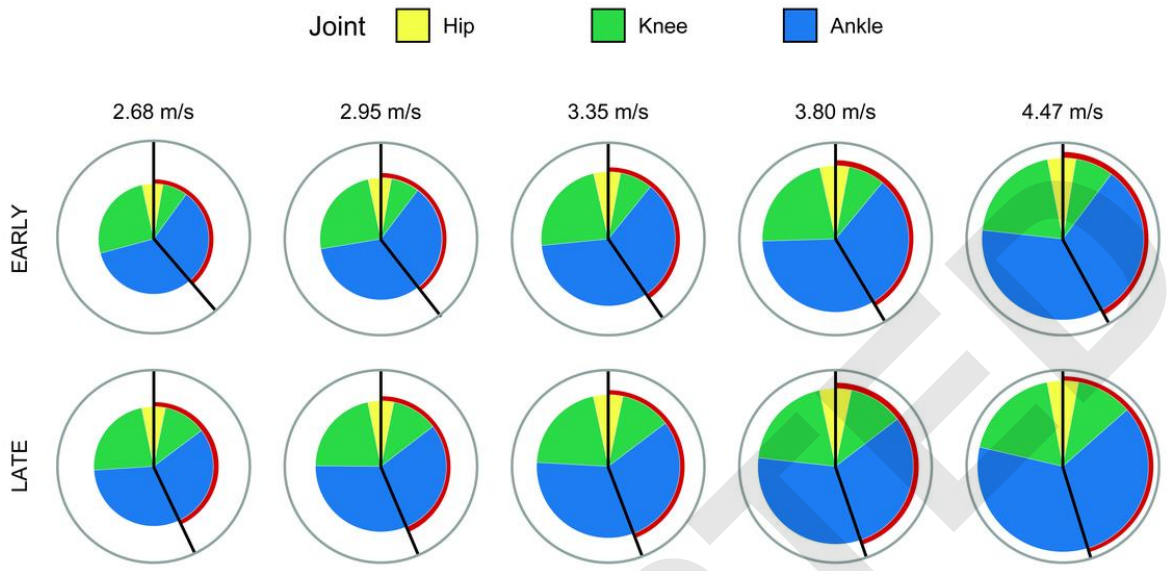

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Figure 4

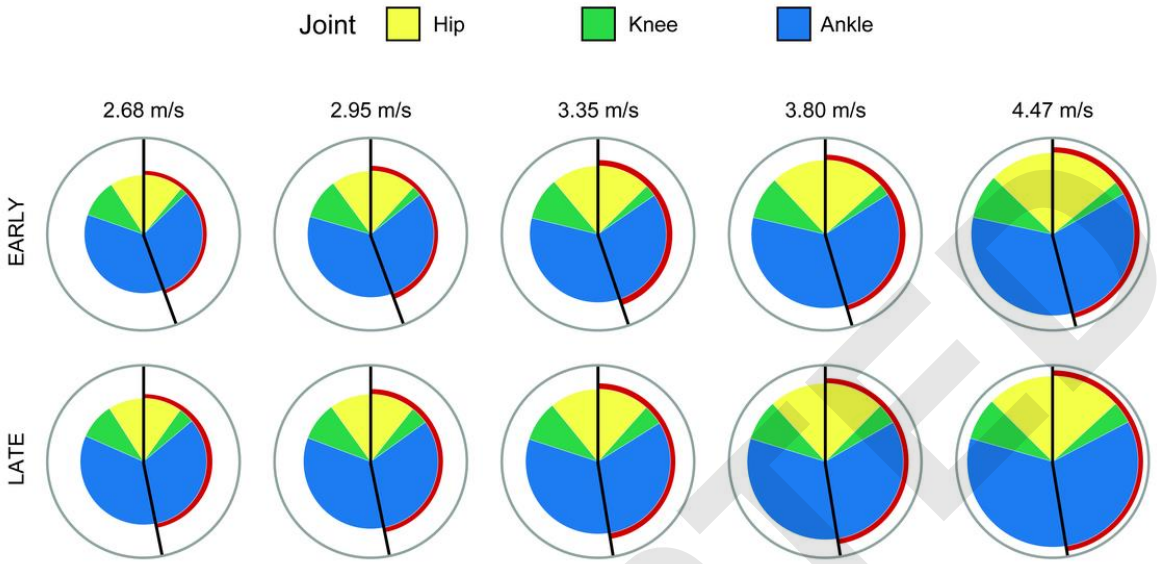

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Table 1. Summary of athlete demographics. There were 56 unique athletes included in this study, of which, 27 completed running assessments at both timepoints. Values are presented as mean (SD) unless otherwise indicated.

	3.5 to 7 Months Post-ACLR	8 to 13 Months Post-ACLR	Overall
N	41	42	56
Age, y	21.0 (1.4)	21.0 (1.5)	21.0 (1.4)
Body Mass, kg	86.1 (18.1)	81.6 (16.4)	84.9 (17.0)
Height, m	1.80 (0.10)	1.78 (0.10)	1.79 (0.10)
Body Mass Index, kg/m^2	26.5 (4.2)	25.6 (3.2)	26.3 (3.8)
Females, n (%)	15 (36.6)	20 (47.6)	24 (42.9)
Graft type, n (%)			
Bone-patellar-tendon-bone	34 (82.9)	36 (85.7)	46 (82.1)
Hamstring tendon	1 (2.4)	3 (7.1)	3 (5.4)
Quadriceps tendon	6 (14.6)	3 (7.1)	7 (12.5)
Concomitant procedures, n (%)			
Meniscectomy	11 (26.8)	10 (23.8)	15 (26.8)
Meniscal repair	12 (29.3)	9 (21.4)	15 (26.8)

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Medial patellofemoral ligament reconstruction	1 (2.4)	1 (2.4)	1 (1.8)
Time post-operatively, months	5.5 (1.0)	9.9 (1.6)	
International Knee Documentation Committee Score[a]	74.4 (10.7)	85.4 (8.8)	
Observations at each speed, n (%)			
2.68 m/s	39 (24.4)	42 (23.9)	
2.95 m/s	38 (23.8)	37 (21.0)	
3.35 m/s	39 (24.4)	40 (22.7)	
3.80 m/s	27 (16.9)	31 (17.6)	
4.47 m/s	17 (10.6)	26 (14.8)	

[a], International Knee Documentation Committee Scores were not available for 3 of the 56 observations.

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Negative Work (J/kg)								
	Hip		**Knee**		**Ankle**		**Total**	
Model Fixed Effects:	**Estimate (95% CI)**	**p-value**	**Estimate (95% CI)**	**p-value**	**Estimate (95% CI)**	**p-value**	**Estimate (95% CI)**	**p-value**
Limb [a]	-0.01 (-0.02, -0.002)	0.025	-0.34 (-0.37, -0.31)	<0.001	-0.05 (-0.08, -0.03)	0.001	-0.42 (-0.47, -0.38)	<0.001
Time [b]	0.002 (-0.01, 0.01)	0.667	-0.02 (-0.03, -0.01)	0.004	0.02 (-0.001, 0.03)	0.065	0.00 (-0.02, 0.02)	0.999
Speed [c]		**< 0.001**		**< 0.001**		**< 0.001**		**< 0.001**
2.95 m/s	0.006 (0.0, 0.01)	0.053	0.02 (0.01, 0.03)	0.002	0.05 (0.04, 0.07)	<0.001	0.10 (0.08, 0.12)	<0.001
3.35 m/s	0.016 (0.01, 0.02)	<0.001	0.04 (0.03, 0.05)	<0.001	0.11 (0.10, 0.12)	<0.001	0.20 (0.18, 0.22)	<0.001
3.80 m/s	0.021 (0.015, 0.028)	<0.001	0.06 (0.05, 0.08)	<0.001	0.17 (0.15, 0.18)	<0.001	0.30 (0.28, 0.32)	<0.001
4.47 m/s	0.019 (0.012, 0.027)	<0.001	0.07 (0.06, 0.08)	<0.001	0.26 (0.25, 0.28)	<0.001	0.43 (0.40, 0.45)	<0.001
Limb [a] * Time [b]	0.012 (0.003, 0.022)	0.010	0.12 (0.10, 0.14)	<0.001	0.02 (0.001, 0.05)	0.039	0.15 (0.12, 0.18)	<0.001
Positive Work (J/kg)								
	Hip		**Knee**		**Ankle**		**Total**	

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Model Fixed Effects:	Estimate (95% CI)	p-value	Estimate (95% CI)	p-value	Estimate (95% CI)	p-value	Estimate (95% CI)	p-value
Limb [a]	0.23 (0.20, 0.26)	<0.001	-0.21 (-0.24, -0.19)	<0.001	-0.12 (-0.15, -0.08)	0.001	-0.28 (-0.34, -0.23)	<0.001
Time [b]	0.01 (-0.00, 0.03)	0.11	-0.01 (-0.02, -0.00)	0.019	0.03 (0.01, 0.05)	0.006	0.03 (-0.003, 0.06)	0.07
Speed [c]		**< 0.001**		**< 0.001**		**< 0.001**		**< 0.001**
2.95 m/s	0.04 (0.03, 0.06)	<0.001	0.02 (0.01, 0.03)	0.001	0.03 (0.01, 0.05)	0.002	0.09 (0.06, 0.12)	<0.001
3.35 m/s	0..09 (0.07, 0.11)	<0.001	0.031 (0.02, 0.04)	<0.001	0.07 (0.06, 0.09)	<0.001	0.20 (0.17, 0.22)	<0.001
3.80 m/s	0.15 (0.13, 0.16)	<0.001	0.033 (0.02, 0.04)	<0.001	0.14 (0.12, 0.16)	<0.001	0.32 (0.29, 0.35)	<0.001
4.47 m/s	0.21 (0.20, 0.23)	<0.001	0.04 (0.03, 0.05)	<0.001	0.22 (0.20, 0.24)	<0.001	0.47 (0.44, 0.51)	<0.001
Limb [a] * Time [b]	-0.02 (-0.04, -0.00)	0.046	0.07 (0.06, 0.09)	<0.001	0.07 (0.04, 0.10)	<0.001	0.12 (0.07, 0.16)	<0.001

Table 2. Overall main effects and fixed effect estimates, including all significant 2-way interactions, from linear mixed effects models assessing the influence of time post-anterior cruciate ligament reconstruction, limb, and speed on hip, knee, ankle, and total negative and positive work during running. [a], the non-surgical limb is the reference for the main effect of limb. [b], the EARLY (3.5-7 months post-operative) group is the reference for the main effect of time. [c], the 2.68 m/s collection is the reference for the main effect of speed. CI, confidence interval; *, 2-way interaction.

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Table S1. Least-square means and 95% confidence interval (CI) for the surgical and non-surgical lower extremity joint and total negative work for all speeds at EARLY (3.5-7-months) and LATE (8-13 months) time points.

	Time (months)	Limb	2.68 m/s	2.95 m/s	3.35 m/s	3.80 m/s	4.47 m/s
Negative Work			**Mean (95% CI)**	**Mean (95% CI)**	**Mean (95% CI)**	**Mean (95% CI)**	**Mean (95% CI)**
Hip	EARLY	Surgical	-0.05 (-0.7, -0.3)	-0.06 (-0.07, -0.04)	-0.07 (-0.08, -0.05)	-0.07 (-0.09, -0.05)	-0.07 (-0.09, -0.05)
		Non-Surgical	-0.06 (-0.08, -0.05)	-0.07 (-0.09, -0.05)	-0.08 (-0.10, -0.05)	-0.08 (-0.10, -0.07)	-0.08 (-0.10, -0.06)
	LATE	Surgical	-0.06 (-0.08, -0.05)	-0.07 (-0.09, -0.05)	-0.08 (-0.10, 0.06)	-0.09 (-0.10, -0.07)	-0.08 (-0.10, -0.06)
		Non-Surgical	-0.06 (-0.08, -0.05)	-0.07 (-0.09, -0.05)	-0.08 (-0.10, 0.06)	-0.09 (-0.10, -0.07)	-0.08 (-0.10, -0.07)
Knee	EARLY	Surgical	-0.13 (-0.17, -0.10)	-0.15 (-0.19, -0.11)	-0.17 (-0.21, -0.14)	-0.20 (-0.23, -0.16)	-0.20 (-0.24, -0.16)
		Non-Surgical	-0.47 (-0.51, -0.43)	-0.49 (-0.53, -0.45)	-0.51 (-0.55, -0.48)	-0.53 (-0.57, -0.50)	-0.54 (-0.58, -0.50)
	LATE	Surgical	-0.23 (-0.27, -0.19)	-0.25 (-0.29, -0.21)	-0.27 (-0.31, -0.23)	-0.29 (-0.33, -0.25)	-0.30 (-0.34, -0.26)
		Non-Surgical	-0.45 (-0.49, -0.41)	-0.47 (-0.51, -0.43)	-0.49 (-0.53, -0.46)	-0.51 (-0.55, -0.48)	-0.52 (-0.56, -0.48)
Ankle	EARLY	Surgical	-0.44 (-0.50, -0.38)	-0.50 (-0.56, -0.44)	-0.55 (-0.61, -0.49)	-0.61 (-0.67, -0.55)	-0.71 (-0.77, -0.64)

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		Non-Surgical	-0.50 (-0.56, -0.43)	-0.55 (-0.61, -0.49)	-0.61 (-0.67, -0.54)	-0.66 (-0.73, -0.60)	-0.76 (-0.82, -0.70)
	LATE	Surgical	-0.48 (-0.54, -0.42)	-0.54 (-0.60, -0.48)	-0.59 (-0.65, -0.53)	-0.65 (-0.71, -0.59)	-0.74 (-0.81, -0.68)
		Non-Surgical	-0.51 (-0.57, -0.45)	-0.57 (-0.63, -0.51)	-0.62 (-0.68, -0.56)	-0.68 (-0.74, -0.62)	-0.77 (-0.84, -0.71)
Total	EARLY	Surgical	-0.71 (-0.76, -0.66)	-0.80 (-0.85, -0.75)	-0.90 (-0.96, -0.85)	-1.01 (-1.06, -0.95)	-1.14 (-1.19, -1.08)
		Non-Surgical	-1.13 (-1.18, -1.08)	-1.23 (-1.28, -1.17)	-1.33 (-1.38, -1.28)	-1.43 (-1.48, -1.38)	-1.56 (-1.61, -1.51)
	LATE	Surgical	-0.86 (-0.91, -0.81)	-0.95 (-1.01, -0.90)	-1.06 (-1.11, -1.00)	-1.16 (-1.21, -1.11)	-1.29 (-1.34, -1.23)
		Non-Surgical	-1.13 (-1.18, -1.08)	-1.23 (-1.28, -1.17)	-1.33 (-1.38, -1.28)	-1.43 (-1.48, -1.38)	-1.56 (-1.61, -1.51)

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Table S2. Least-square means and 95% confidence interval (CI) for the surgical and non-surgical lower extremity joint and total positive work for all speeds at EARLY (3.5-7-months) and LATE (8-13 months) time points.

	Time (months)	Limb	2.68 m/s	2.95 m/s	3.35 m/s	3.80 m/s	4.47 m/s
Positive Work			**Mean (95% CI)**	**Mean (95% CI)**	**Mean (95% CI)**	**Mean (95% CI)**	**Mean (95% CI)**
Hip	EARLY	Surgical	0.28 (0.24, 0.31)	0.32 (0.29, 0.35)	0.37 (0.33, 0.40)	0.42 (0.39, 0.45)	0.49 (0.46, 0.52)
		Non-Surgical	0.23 (0.20, 0.26)	0.27 (0.24, 0.31)	0.32 (0.29, 0.35)	0.38 (0.34, 0.41)	0.44 (0.41, 0.48)
	LATE	Surgical	0.27 (0.23, 0.30)	0.31 (0.28, 0.34)	0.36 (0.33, 0.39)	0.41 (0.38, 0.45)	0.48 (0.45, 0.51)
		Non-Surgical	0.24 (0.21, 0.28)	0.29 (0.25, 0.32)	0.34 (0.30, 0.37)	0.39 (0.36, 0.42)	0.46 (0.42, 0.49)
Knee	EARLY	Surgical	0.05 (0.03, 0.07)	0.07 (0.04, 0.09)	0.08 (0.06, 0.10)	0.08 (0.06, 0.11)	0.09 (0.07, 0.12)
		Non-Surgical	0.26 (0.24, 0.29)	0.28 (0.26, 0.30)	0.29 (0.27, 0.32)	0.30 (0.27, 0.32)	0.31 (0.28, 0.33)
	LATE	Surgical	0.11 (0.08, 0.13)	0.12 (0.10, 0.15)	0.14 (0.11, 0.16)	0.14 (0.12, 0.17)	0.15 (0.13, 0.18)
		Non-Surgical	0.25 (0.23, 0.27)	0.27 (0.24, 0.29)	0.28 (0.26, 0.31)	0.28 (0.26, 0.31)	0.29 (0.27, 0.32)
Ankle	EARLY	Surgical	0.79 (0.75, 0.83)	0.82 (0.78, 0.86)	0.86 (0.82, 0.91)	0.93 (0.89, 0.97)	1.01 (0.96, 1.05)

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		Non-Surgical	0.91 (0.86, 0.95)	0.94 (0.89, 0.98)	0.98 (0.94, 1.02)	1.04 (1.00, 1.09)	1.12 (1.08, 1.17)
	LATE	Surgical	0.89 (0.84, 0.93)	0.92 (0.87, 0.96)	0.96 (0.92, 1.00)	1.02 (0.98, 1.07)	1.10 (1.06, 1.15)
		Non-Surgical	0.94 (0.89, 0.98)	0.96 (0.92, 1.01)	1.01 (0.97, 1.05)	1.07 (1.03, 1.12)	1.15 (1.11, 1.20)
Total	EARLY	Surgical	1.12 (1.06, 1.17)	1.20 (1.15, 1.26)	1.31 (1.26, 1.37)	1.43 (1.38, 1.49)	1.59 (1.53, 1.65)
		Non-Surgical	1.40 (1.35, 1.45)	1.49 (1.43, 1.54)	1.60 (1.54, 1.65)	1.72 (1.66, 1.77)	1.87 (1.82, 1.93)
	LATE	Surgical	1.26 (1.21, 1.31)	1.35 (1.29, 1.40)	1.46 (1.40, 1.51)	1.58 (1.52, 1.63)	1.74 (1.68, 1.79)
		Non-Surgical	1.43 (1.38, 1.48)	1.52 (1.46, 1.57)	1.62 (1.57, 1.68)	1.75 (1.69, 1.80)	1.90 (1.85, 1.96)

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